Dr. ir. Troy Nachtgall

Personalized Clothing and Individual Equiptment (CIE) for Dynamic Interaction and Obfuscation

Wearable Senses Lab, Textile Ecosystem Digital Fieldlab

Thermodynamic Camaflouge

Additive Manufacturing (3D Printing and 3D Textiles) for thermal systems that absorb heat and obstruct IR camera technologies.

Sensing and thermodynamics

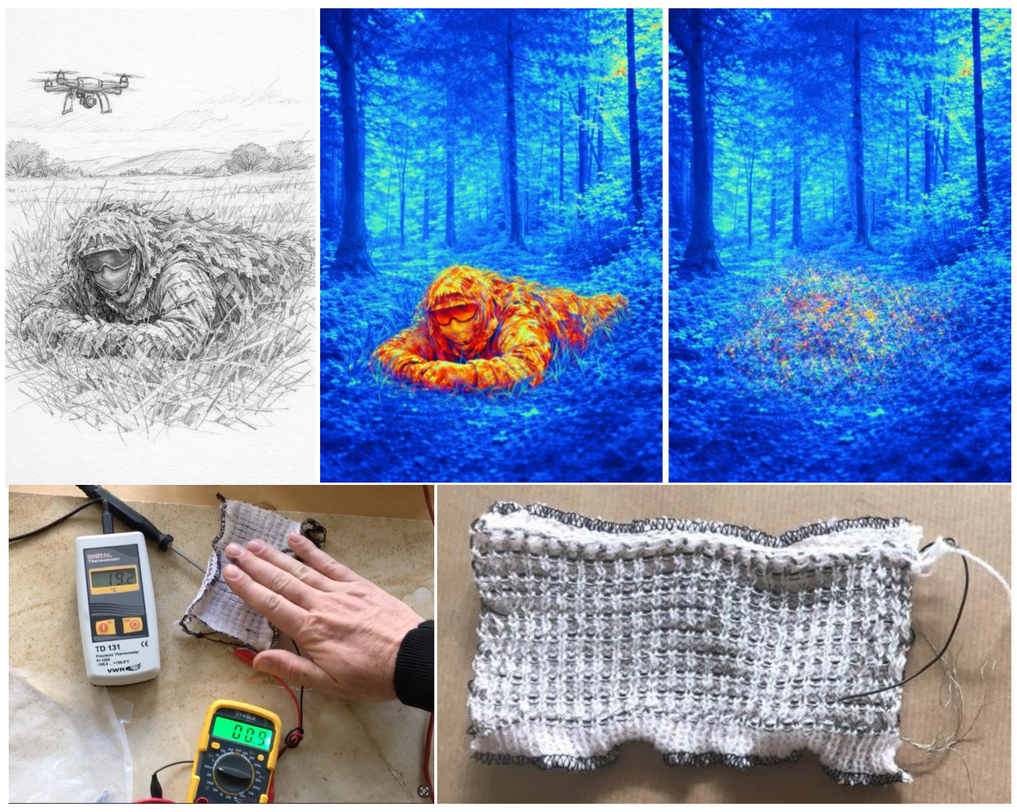
• Integrated thermoelectric structures using the Peltier–Seebeck effect enable localized heating, cooling, and heat redistribution
• Textile-integrated systems regulate personal thermal comfort during prolonged deployment
• Thermodynamic control can modulate external heat signatures to reduce infrared detectability
• Distributed secondary sensors detect impact events and environmental thermal changes

Fabrication and computational structures

• Digitally fabricated structures embedded in flexible substrates
• Electronic and mechanical computation integrated directly into material structures
• Structures support distributed sensing, thermal management, and energy transfer within the system

Interaction and deployment dynamics

• Interaction design supporting continuous movement
• Systems maintain functionality under sustained motion, load transfer, and environmental stress
• Thermodynamic and sensing feedback adapts in real time to activity level and operational context



Personalized Dynamic CIE

Sensor and Actuator systems for enhanced ability and protection in Clothing and Individual Equiptment (CIE)

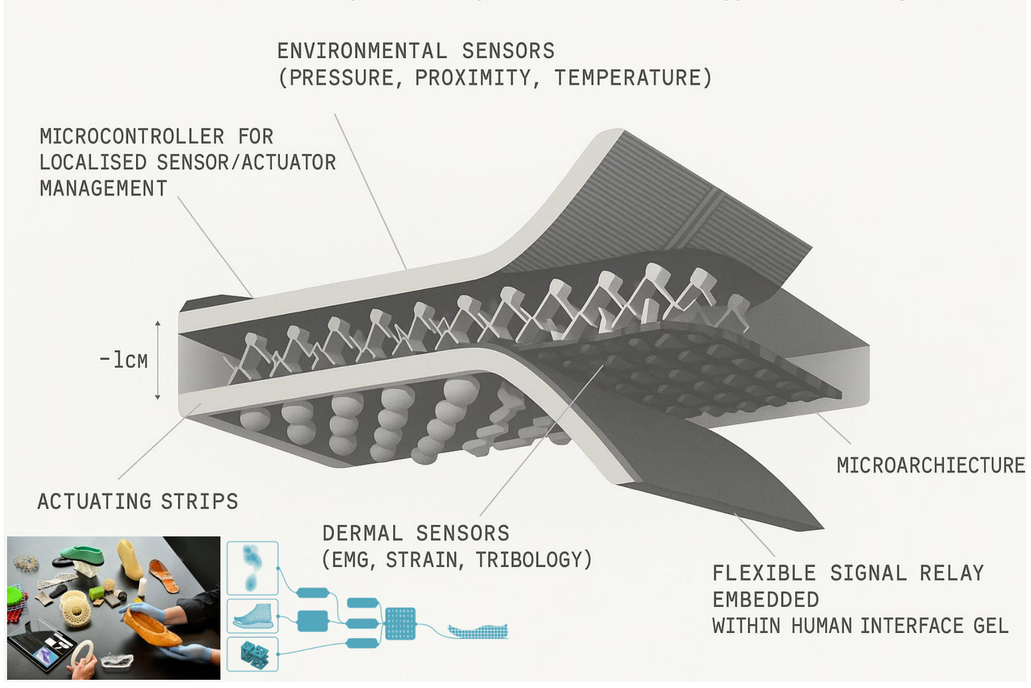
Collaboration with Dr. Miguel Dias of Biomedical Engineering

Sensing and communication layer

•Bio-printed electronic sensor measure load, joint angles, temperature, moisture, and biochemical markers
•Active sensing streams data to edge-AI nodes detecting fatigue, injury onset, and thermal strain
•Low-profile textile haptics (vibration, pneumatic pouches) provide screenless communication and feedback
•Passive bio indicators (color, odor) reveal biomarkers for situation like infection risks over long-term use

Movement augmentation layer

•Integrated actuators act as artificial muscles through precision additive manufacturing for personalized interaction
•Soft exomuscle layer enhance specific nerve and muscle groups
•Edge-AI control learns gait and fatigue patterns to deliver just-in-time assistance
•Continuous sensing builds a biomechanical model for training, load management, and recovery

Protective outer layer

•Thin engineered armor combining aramid and UHMWPE textiles
•3D-printed auxetic lattices and hydrogels distribute and absorb kinetic impact
•Anatomical scanning enables individualized stiffness and thickness
•Embedded sensors capture impact data for triage and analysis